

DS Lab 9: Data Visualization II

Box Plot on Titanic Dataset

Aim:

Plot a box plot for distribution of age with respect to gender and survival status in the Titanic dataset.

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_style('whitegrid')
```

```
In [2]: dataset = sns.load_dataset('titanic')
dataset.head()
```

```
Out[2]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

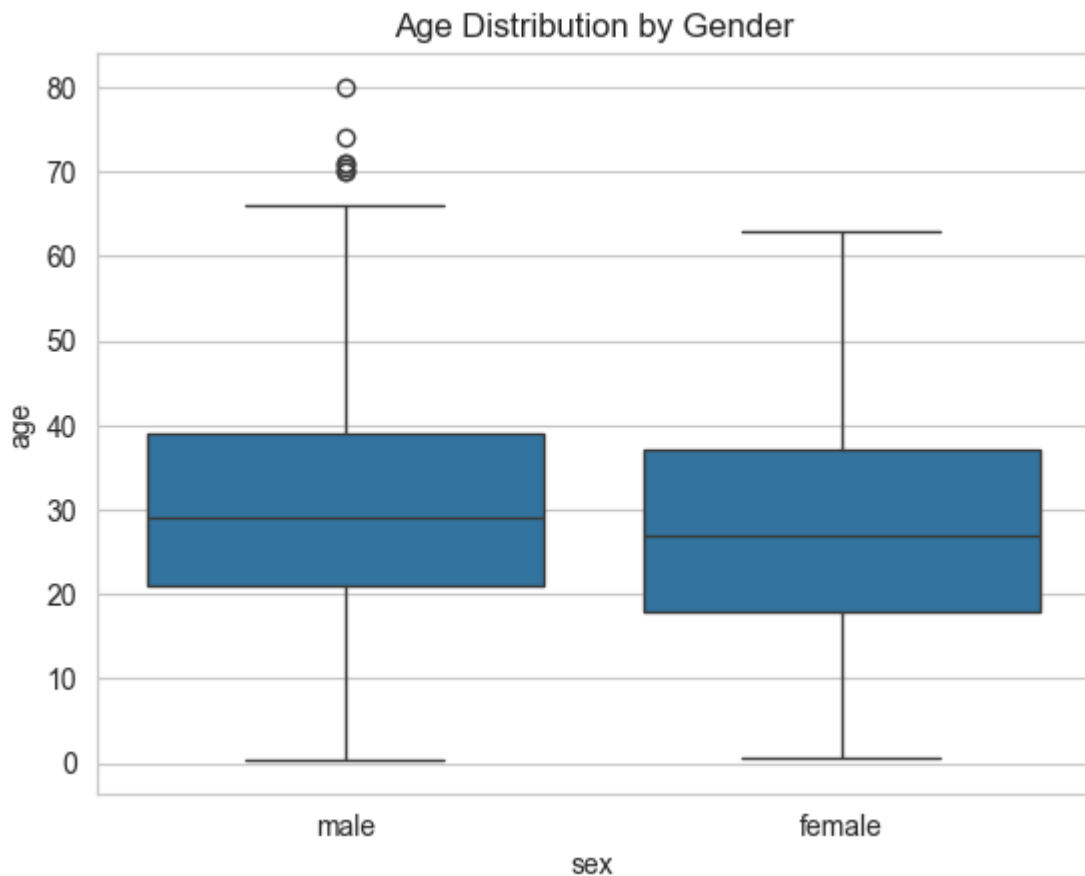


```
In [3]: dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male   891 non-null    bool
11  deck        203 non-null    category
12  embark_town  889 non-null    object
13  alive        891 non-null    object
14  alone       891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

Box Plot: Age vs Gender

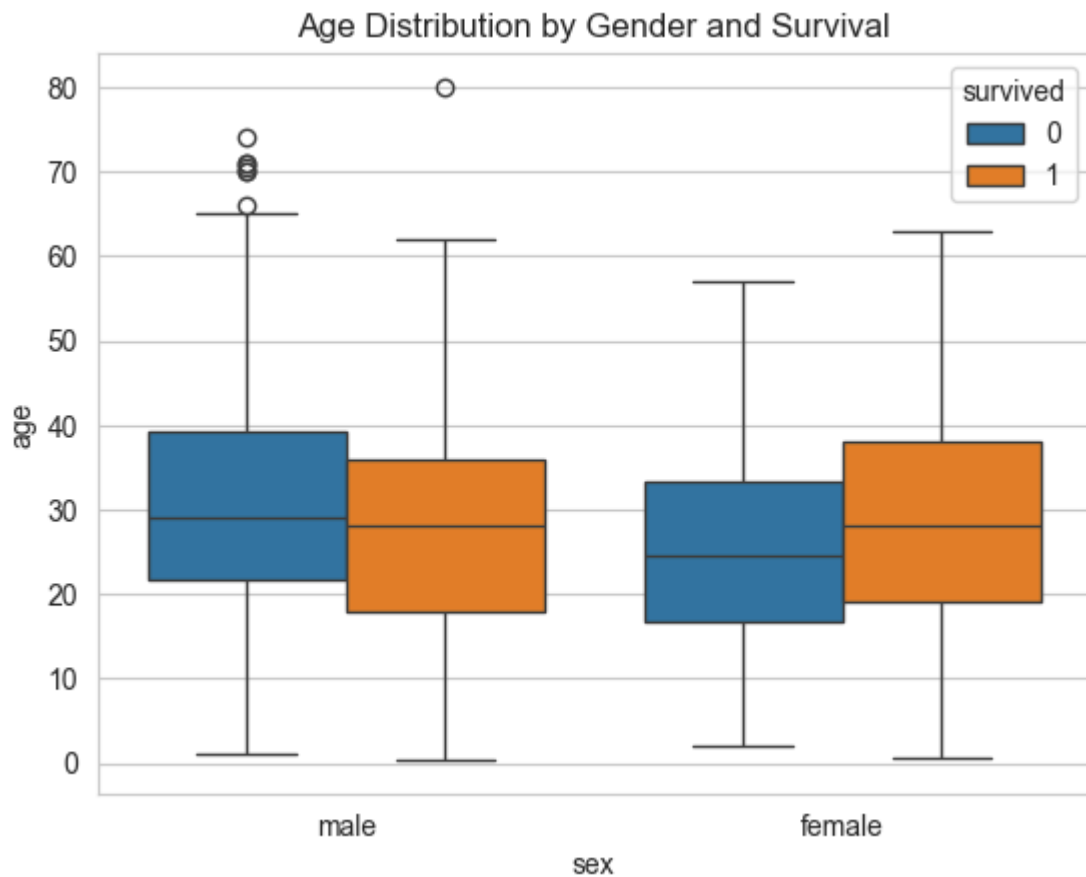
```
In [4]: sns.boxplot(x='sex', y='age', data=dataset)
plt.title("Age Distribution by Gender")
plt.show()
```



Insight: Male passengers show wider age variation compared to female passengers.

Box Plot: Age vs Gender and Survival

```
In [5]: sns.boxplot(x='sex', y='age', data=dataset, hue='survived')  
plt.title("Age Distribution by Gender and Survival")  
plt.show()
```



Insight: Younger passengers, especially females, had higher survival rates compared to older passengers.

Conclusion

The box plot visualizes age distribution across gender and survival categories, highlighting median differences and outliers.